

Alexandre AUTRAN

Mathematics and Machine Learning Student

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Profile

I am a mathematics and machine learning student (*élève normalien*) at ENS Rennes and ENS Paris-Saclay. My research interests include high-dimensional statistics, stochastic processes, optimal transport, and optimal control. I combine theoretical analysis with numerical methods and machine learning implementations in Python, with a current interest in graph learning and generative models.

Education

2024–2027 Normalien – ENS Diploma in Mathematics

Admitted to the ENS through a competitive entrance examination.
Four-year diploma pursued over three years (accelerated track).

2026–2027 **Research Year in Machine Learning**, *École Normale Supérieure Paris-Saclay*, France
Final year of the ENS Diploma (research track, “Prelab” ENS).

2025–2027 **MSc in Mathematics**, *École Normale Supérieure de Rennes*, France
M2 Fundamental Mathematics. Highest honors (17/20 in the first semester, completed alongside M1).
M1 Fundamental Mathematics. Highest honors (17.6/20). Rank: 5/100+.

2024–2025 **BSc in Mathematics, minor in Computer Science**, *École Normale Supérieure de Rennes*, France
L3 Fundamental Mathematics. High honors (15.4/20).

2022–2024 **BSc in Mathematics, minor in Computer Science**, *University of Rennes*, France
L2 Mathematics. Highest honors (18/20). Rank: 3/100+.
L1 Mathematics. Highest honors (18.5/20). Rank: 1/100+.

Publications

[\[More details\]](#)

- [1] **A. Autran**, Conditional Fluctuations of Mean-Field Systems with Stable Common Noise, preprint, 2026.
Under review at *Electronic Journal of Probability*.
- [2] **A. Autran**, Singular Tracking of Fractional Brownian Motion under Proportional Costs, preprint, 2026.
Submitted to *Stochastic Processes and their Applications*.
- [3] **A. Autran** et al., Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices
Subjected to Single Ionizing Particles, *Applied Sciences*, 16(11), 5551, 2026.

Research Experience

[\[More details\]](#)

Sep 2026–May 2027 Machine Learning Internship

High-Dimensional Density Analysis Using Local Unimodality.
Supervised by Argyris Kalogeratos (Senior Research Scientist) at Centre Borelli, ENS Paris-Saclay, France.

May–Aug 2026 M1 Research Internship

Optical Modeling of VO₂-Based Thermochromic Oxide Nanocomposites.
Supervised by Nguyen Ngoc Duy (Prof.) and Amaury Baret (PhD Student) at the University of Liège, Belgium.

Apr–May 2026 Research Internship

Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems.
Supervised by Martin Gander (Prof.) and Bastien Chaudet-Dumas (Postdoctoral Researcher) at the University of Geneva, Switzerland.

Jun–Aug 2025 L3 Research Internship

Lattice-Based Post-Quantum Cryptography.
Supervised by Pierre-Alain Fouque (Prof.) and Aurore Guillevic (Research Scientist) at Inria, France.

May–Jun 2025 Research Internship

Algebraic Counting of the Real Roots of Polynomials.
Supervised by Marie-Françoise Roy (Prof. Emeritus) at the University of Rennes, France.

GitHub Projects

- 2026 **Comparing Dense and Sparse Neural Models for Constrained Graph Generation**
PyTorch implementation of diffusion-based graph generation with hard connectivity and edge-budget guarantees, with a focus on structural validity and efficient generation.
- 2026 **Optimal Control of Spatial Lotka–Volterra Systems**
Numerical simulation and constrained optimization of spatial Lotka–Volterra reaction–diffusion systems using an exact discrete adjoint.

Other Activities

- 2024–2025 **ENS Oral Examinations Preparation**
Supervision of second-year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures (Ulm, Paris-Saclay, Rennes, and Lyon).
- 2022–2025 **Teaching in Mathematics**
Tutoring for final-year high school students at Lycée Chateaubriand in Rennes, and for first- and second-year BSc students at the University of Rennes.
- 2023–2024 **BSc Student Representative**
Representative on the improvement council of the Mathematics Department of the University of Rennes.
- Before 2022 **High School Competitions**
Participation in the Concours Général in Mathematics, Physics, and Philosophy, and the National High School Mathematics Olympiad, at Lycée Thiers in Marseille.

Technical Skills

Machine Learning: PyTorch, scikit-learn.
Scientific Computing: Python (NumPy, pandas), MATLAB, R.
Programming: Python, C/C++, Java.
Tools: Git, $\text{\LaTeX}$.
Methods: numerical optimization, optimal control, probabilistic methods.

Languages

French: native. **English:** professional working proficiency.

Sports and Interests

Sports: running, cycling, strength training.
Interests: fantasy and science fiction literature; chess.

Academic References

François Bolley — francois.bolley@ens-rennes.fr
Full Professor, IRMAR, ENS de Rennes.
Director of the Mathematics Department at ENS de Rennes.

Jean-Christophe Breton — jean-christophe.breton@univ-rennes.fr
Full Professor, IRMAR, University of Rennes.

Françoise Dal’Bo — francoise.dalbo@univ-rennes.fr
Full Professor, IRMAR, University of Rennes.
Director of the International Research Laboratory IFUMI.

Martin Gander — martin.gander@unige.ch
Full Professor, Section of Mathematics, University of Geneva.

Argyris Kalogeratos — argyris.kalogeratos@ens-paris-saclay.fr
Senior Research Scientist, Centre Borelli, ENS Paris-Saclay.